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# Building Grouping and Soft Actor Routing in MAPPO for CityLearn Energy Control

Progress Report 2

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# Chapter 1

## Introduction

Modern homes are no longer only passive electricity consumers. They may include solar panels, batteries, heating and cooling equipment, and other controllable devices. Coordinating these resources across a community can move electricity use away from peak periods and make the combined load follow a desired grid profile. The difficulty is that each building has different demand and flexibility, while control performance is evaluated for the whole portfolio.

Building energy research has therefore increasingly moved from fixed rule-based control toward reinforcement learning and multi-agent coordination [3, 19]. This project uses CityLearn, an open simulation environment that provides a standardized and reproducible way to compare building-control algorithms [16, 11]. The experiment follows the IEA EBC Annex 96 Common Exercise 1 benchmark: 25 single-family homes are controlled as one portfolio to track a daily reference load profile using battery and HVAC flexibility. Evaluation covers CV-RMSE, NMBE, indoor comfort (measured as temperature deviation multiplied by duration and averaged per building per test day), energy use, peak demand, ramping, load factor, and fairness. Vermont uses January for training and February for testing, while Texas uses August and September.

The first progress report, completed prior to this report, established a reproducible CityLearn experimental pipeline based on the IEA EBC Annex 96 Common Exercise 1 (CE1) benchmark. It compared rule-based control, independent reinforcement learning, standard MAPPO, grouped MAPPO, and several communication modules for a portfolio of 25 homes. The main result was that grouped cooperative control was substantially stronger than the fixed rule-based baseline, while the choice of communication structure materially changed load tracking and thermal com-

fort. Among the tested communication directions, TarMAC showed the greatest potential. The previous report also identified several unresolved problems: the original TarMAC communication method did not fully use each agent’s own information, and K-means grouping was treated as a fixed preprocessing step rather than an independent research variable. Preliminary experiments showed that K-means produced weak groups, which directly affected the subsequent MAPPO training. The current MAPPO model depends strongly on the quality of the fixed grouping, and this quality depends on both the grouping method and the selected grouping features.

The core research question of this second report is:

In the CityLearn portfolio load-tracking task, how should communication, building grouping methods, and grouping features be designed to improve MARL? Alternatively, is there a method that can reduce the model’s dependence on a fixed grouping?

**The main innovation of this work** is a Soft Router for grouped MAPPO that dynamically selects a complete actor for each building according to its current state, relaxing the permanent building-to-actor assignment imposed by fixed grouping.

This second report addresses those problems in four connected studies. First, it introduces *TarMAC Hybrid*. It reduces each agent’s own information to a smaller and clearer form and uses ReLU to keep the more useful parts. This shortened information is then combined with information received from other agents, while the original information is retained. Second, the report compares K-means, Gaussian mixture models (GMM), Ward agglomerative clustering, and a balanced spectral method. Third, it studies both the number and composition of grouping features and ultimately identifies a compact three-feature design, denoted 3fA: battery energy-storage capacity, mean heating demand, and mean non-shiftable load. Finally, it develops a soft router. In simple terms, this is a small neural-network selector for each building. At every time step, it examines the building’s current condition and gives each available actor a selection probability. These probabilities help decide which actor should guide the building’s action. This tests whether a building can use different actors at different times instead of always depending on the actor assigned by its fixed group.

Chapter 2 reviews the literature needed for the new grouping, feature-selection, and routing studies. Chapter 3 presents the implemented grouping-method comparison, feature selection, and the evolution of the soft router. Chapter 4 reports the experiments using the consolidated re-

sult tables in `grouping_feature_ablation_primary_only_sorted` and `summarize_selected_experiment_metrics`.

Chapter 5 summarizes the conclusions and remaining work.

# Chapter 2

## Literature Review

### 2.1 Building Energy Coordination and CityLearn

CityLearn was designed to standardize reinforcement-learning research for demand response and urban energy management, and its later version extends the environment to cover communities that consider both grid needs and occupant comfort [16, 11]. This is useful here because the controller must manage several kinds of building information at the same time. Battery capacity tells us how much energy a building can store and move to a different time. Heating or cooling demand tells us how much HVAC energy may need to be controlled. Non-shiftable load is the ordinary electricity use that cannot be moved by the controller. Recent residential MARL studies also focus on coordinating many flexible homes while allowing each home to make its own control decision [3, 19].

In a power grid, one control action can affect several building results at the same time. For example, charging or discharging a battery changes the building’s electricity load. Changing HVAC operation affects both electricity use and indoor temperature. When many homes act at the same time, their loads add together and can change the community peak or make the total load rise and fall more quickly. Research on power networks shows that adding known physical information can help MARL make better control decisions [20]. Research on safe building control also shows that a small energy error does not automatically mean that comfort and network limits are satisfied [17, 15]. These studies suggest that buildings should not be grouped using only one measure of electricity use. This project therefore considers battery capacity,

which affects how much load can be moved to another time; heating demand, which reflects the building’s heating needs; and non-shiftable load, which is the ordinary electricity use that the controller cannot move. Using these features allows buildings with similar energy use and similar control tasks to be placed in the same group.

## **2.2 Grouping Algorithms**

### **2.2.1 K-means**

In the previous experiments, K-means was used as the initial grouping baseline because it is simple and fast [8]. It also reduced the overall computational complexity, which was helpful for the later study of communication methods. K-means first places several group centres, assigns every building to its nearest centre, updates the centres, and repeats these steps. In plain terms, it tries to put buildings with similar feature values into the same group. However, if the starting centres change from one run to another, the final result may also change. It works best for simple, compact groups and does not guarantee similar numbers of buildings in every group. Large differences in group size can seriously affect the later training: an actor for a small group receives less training experience than an actor for a large group and can be more easily influenced by information from the large group. Therefore, K-means is not a sufficiently stable method for the overall experiment.

### **2.2.2 Why These Alternative Methods Were Selected**

Strictly speaking, this task requires clustering rather than supervised classification. There is no given label that says which actor group is correct for each building; only the building features are available. The selected methods must therefore discover groups without labels, assign every building to one group.

GMM, agglomerative clustering, and balanced spectral clustering were selected because they satisfy these requirements while examining three different ways of forming groups. GMM tests whether allowing overlapping groups with different spreads improves on K-means. Agglomerative clustering tests a deterministic step-by-step merging process and avoids random starting



centres. Balanced spectral clustering uses pairwise building similarities and directly prevents very unequal group sizes. GMM and hierarchical clustering have also been used together to identify electricity-use patterns across multiple buildings, which makes them relevant to this application [7]. The three methods therefore provide complementary comparisons without greatly expanding an experiment that must later train several reinforcement-learning models.

Many other classification methods, such as xRFM, XGBoost, and Random Forest, were not selected because they answer a different type of question. They are supervised prediction methods: training examples must include a known output, such as a class label or regression target, before the model can learn what to predict [1, 4, 2]. This project has no correct actor-group labels. Creating labels with another clustering method and then training one of these predictors would mainly teach it to copy the earlier clustering result, rather than discover a better grouping. Some unsupervised alternatives, such as density-based clustering, can choose a variable number of groups or leave some buildings unassigned as noise. That is also inconvenient here because every building must be assigned to an actor and the experiments compare fixed group counts. These methods may be useful for a different prediction task or a larger dataset, but they do not match the main question of the present grouping experiment as directly.

### 2.2.3 Gaussian Mixture Model

A Gaussian mixture model (GMM) treats the data as if they were produced by several overlapping bell-shaped groups [6]. Instead of only measuring the distance to a centre, GMM estimates the probability that a building belongs to each group. The building can then be assigned to the group with the highest probability. This allows GMM to describe groups that have different widths, directions, or some overlap, so it is more flexible than K-means. The disadvantage in this experiment is the small dataset: there are only 25 buildings. A full GMM must estimate many values from these 25 examples, so its grouping can be unstable or too strongly influenced by a few buildings.

### 2.2.4 Agglomerative Clustering

Agglomerative clustering works from small groups to large groups. At the beginning, every building is its own group. The method then repeatedly merges the two most similar groups until the required number of groups remains. This project uses Ward’s rule, which chooses the merge that keeps the buildings inside each new group as similar as possible [18]. The method does not need random starting centres, so the same input produces the same grouping. For a small experiment with only 25 buildings, this reduces the effect of random grouping on the later MAPPO experiment. Its limitation is that an early merge cannot be undone, and it does not guarantee exactly equal group sizes.

### 2.2.5 Balanced Spectral Clustering

Spectral clustering first builds a similarity graph [10]. Each building is a point in the graph, and two buildings receive a stronger connection when their selected features are more similar. The method then uses the main patterns of this graph to create a simpler representation in which strongly connected buildings are easier to separate from the others. This step is why the method is called “spectral”: it studies numerical patterns in the similarity graph. The buildings are then grouped using this simpler view of their relationships.

The balanced version used in this project adds one more rule: group sizes must be equal or differ by at most one building. It creates a fixed number of places in each group. It then uses a standard matching method, called the Hungarian assignment algorithm, to place all buildings while keeping the total difference between buildings and their assigned groups as small as possible. This follows the idea of balanced clustering [9]. This prevents one actor from receiving many buildings while another actor receives very few. Unlike K-means, spectral clustering does not require all buildings in a group to gather around one centre. It can keep a chain of buildings in the same group when each building is similar to the next, even if the buildings at the two ends are not very close.

In this experiment, “similarity” is not an additional building feature. The selected features are first standardized so that their scales are comparable. The method then calculates the overall distance between each pair of buildings and uses an RBF rule to convert that distance into a

connection strength: a shorter distance gives a stronger connection, while a longer distance gives a weaker connection. Therefore, changing the selected features or the rule used to convert distance into connection strength can change the grouping result. Forcing equal group sizes may also separate buildings that would otherwise be placed together.

## 2.3 Grouping Features and Redundancy

Using more features does not always produce better groups. A useful feature should help distinguish buildings, but it should not simply repeat information that is already present. This idea is often described as selecting features that are relevant but not redundant [13]. For example, mean heating demand, maximum heating demand, and HVAC rating all describe related parts of a building’s heating requirement. If all three are included, heating may influence the grouping several times, while battery or ordinary electrical load is counted only once. The feature ablation therefore tests small combinations instead of assuming that every available column should be used.

The Annex 96 benchmark provides 37 related features. For the initial grouping tests, one feature can first be randomly selected from each main part. For example, the best feature set found after testing, referred to here as 3fA, was formed by selecting one feature from each of three main parts: **bes\_capacity\_kwh** tells us how much energy the battery can move, **heating\_mean** describes the average heating need during the Vermont winter, and **ns1\_mean** describes the average ordinary electrical load that the controller cannot move. This does not mean that these three features are best for every problem. Chapter 4 compares different feature counts and combinations and shows that this simple set is more effective than a larger feature set containing repeated information.

## 2.4 Soft Routing and Mixtures of Experts

Mixture-of-experts models use several candidate neural networks and another small network that chooses between them. The candidate networks are often called “experts”, and the choosing network is called a “router”. In soft routing, the router gives every candidate a probability

instead of immediately choosing only one. Candidates with higher probabilities have more influence on the result. Previous Soft MoE and reinforcement-learning studies show that this kind of selection can help different networks learn different jobs [14, 12]. In this project, the candidates are the actors trained for different building groups. For each building and each time step, the router examines the current building information and helps select which actor should guide the action.

The selector is difficult to train when the candidate actors have not learned useful behaviours. In that case, choosing one actor instead of another provides little benefit. Training the actors and the selector together from the beginning can also be unstable because both sides keep changing. The staged experiments therefore first train actors with fixed groups, then keep those actors unchanged while training the selector, and finally test whether a small amount of further actor training is helpful. This means that soft routing is treated as an improvement built on a good initial grouping, rather than a method that is expected to create good groups by itself from the beginning.

# Chapter 3

## Method and Progress

This chapter follows the same grouped MAPPO and centralized-training, decentralized-execution foundation used in the previous report. The new work is concentrated in the actor communication block, the construction of building groups, and the selection of actor experts. This stage also investigates which reward-function weight setting works best. The basic reward form is retained, but several combinations of its weights are compared to study the balance between load tracking and indoor comfort.

### 3.1 Reward Function and Parameter Comparison

At time step  $t$ , let  $y_t$  be the total electricity load of all  $N$  buildings, let  $r_t$  be the target load, and let  $e_t = y_t - r_t$  be the tracking error. The two running load-tracking terms used by the reward are

$$B_t = \left| \frac{\sum_{\tau=1}^t e_{\tau}}{\sum_{\tau=1}^t r_{\tau}} \right|, \quad (3.1)$$

$$E_t = \frac{\sqrt{\frac{1}{t} \sum_{\tau=1}^t e_{\tau}^2}}{\frac{1}{t} \sum_{\tau=1}^t r_{\tau}}, \quad (3.2)$$

where  $B_t$  is the running absolute NMBE term and  $E_t$  is the running CV-RMSE term. NMBE shows whether the total load is generally above or below the target over time: a positive value means that it is generally too high, while a negative value means that it is generally too low. The reward uses its absolute value, so a result closer to zero means less overall bias. CV-RMSE

shows the typical size of the hour-by-hour tracking errors relative to the average target load, and larger errors have a stronger effect on it. A lower CV-RMSE therefore means that the load follows the target more closely. For building  $i$ , the number of degrees outside the seasonal comfort range  $[L_t, U_t]$  and the corresponding violation indicator are

$$d_{i,t} = \max(L_t - T_{i,t}, 0) + \max(T_{i,t} - U_t, 0), \quad (3.3)$$

$$I_{i,t} = \mathbb{I}[d_{i,t} > 0]. \quad (3.4)$$

The complete per-building reward is therefore

$$R_{i,t} = - \left[ \frac{w_{\text{nmbe}} B_t + w_{\text{cv\_rmse}} E_t}{N} + w_{\text{comfort}} (w_{\text{binary}} I_{i,t} + w_{\text{degree}} d_{i,t}) \right]. \quad (3.5)$$

The first two penalties are calculated from the total load and divided equally among the  $N$  buildings. The final term is calculated separately for each building. It gives one penalty when the indoor temperature leaves the comfort range and an additional penalty according to how many degrees it exceeds the range. In the reported experiments, Vermont uses January for training and February for testing, with a comfort range of 20–24°C. Texas uses August for training and September for testing, with a comfort range of 22–26°C.

The five commonly used parameters are the NMBE weight  $w_{\text{nmbe}}$ , CV-RMSE weight  $w_{\text{cv\_rmse}}$ , overall comfort weight  $w_{\text{comfort}}$ , fixed comfort-violation weight  $w_{\text{binary}}$ , and temperature-exceedance weight  $w_{\text{degree}}$ . In this order, the usual baseline setting is (1.0, 1.0, 0.8, 1.3, 0.3). These objectives can conflict. For example, moving battery or HVAC demand to improve indoor comfort can move the portfolio load away from its target, while concentrating only on load tracking can leave less freedom to correct indoor-temperature violations. NMBE and CV-RMSE can also favour different behaviours because one measures overall bias and the other measures hour-by-hour tracking error.

Four reward settings were compared: the baseline above; a stricter-comfort setting (1.0, 1.0, 1.5, 3.0, 1.0); a degree-hours follow-up (1.0, 1.0, 1.5, 3.0, 1.5); and a load-tracking-focused setting (5.0, 5.0, 1.0, 1.0, 1.5). The stricter-comfort setting keeps both load-tracking weights at 1.0 while increasing the three comfort-related weights. The next comparison keeps the first four weights unchanged and increases only  $w_{\text{degree}}$  from 1.0 to 1.5. This controlled change tests whether a larger penalty for the size of a temperature violation actually reduces degree-hours. The load-tracking-focused setting

then tests the opposite direction by increasing the NMBE and CV-RMSE weights. The best setting is chosen from the measured load-tracking and comfort results rather than from reward magnitude alone.

## 3.2 TarMAC Hybrid Communication

The original implementation follows targeted multi-agent communication (TarMAC) [5]. For hidden state  $h_i \in \mathbb{R}^{256}$ , it forms three representations. The query describes what information building  $i$  is looking for. The key describes what information each building can provide and is used for matching. The value contains the actual information to be passed. They are calculated as

$$q_i = W_q h_i, \quad k_i = W_k h_i, \quad v_i = W_v h_i, \quad (3.6)$$

$$a_{ij} = \text{softmax}_j \left( \frac{q_i^\top k_j}{\sqrt{d_k}} \right), \quad c_i = \sum_j a_{ij} v_j, \quad (3.7)$$

The model compares the query of one building with the keys of all buildings. A better match receives a larger weight, and the values are combined according to these weights to form the received communication information  $c_i$ . The division by  $\sqrt{d_k}$  only prevents the matching scores from becoming too large. The original implementation then computes an update directly from the complete local state and  $c_i$ . Because the full 256-dimensional local state enters the fusion layer without compression, repeated local information can overwhelm the 64-dimensional communication information in this 25-building task.

TarMAC Hybrid changes the fusion path to

$$\ell_i = \text{ReLU}(W_\ell h_i), \quad \ell_i \in \mathbb{R}^{64}, \quad (3.8)$$

$$\Delta_i = \tanh(W_2 \tanh(W_1[\ell_i; c_i] + b_1) + b_2), \quad (3.9)$$

$$h'_i = h_i + \Delta_i. \quad (3.10)$$

Compared with the original TarMAC communication method, TarMAC Hybrid makes three main changes. First, a learned  $256 \rightarrow 64$  projection turns the local state into a more compact representation and reduces repeated information. Second, ReLU provides nonlinear filtering so that useful patterns in the current input are more clearly represented. Third, a residual

connection adds the communication result to the original state instead of replacing the original state completely. Together, these changes first simplify each building’s local information and then combine it with information from other buildings without losing the original input. The default ReLU experiment uses 32-dimensional queries and keys and 64-dimensional values. It uses one global communication round, meaning that all 25 buildings exchange information once before the action is produced, and each building can consider information from every other building rather than only buildings in its fixed group.

To determine which part of TarMAC Hybrid produces the improvement, the experiments also compare two alternatives with the default ReLU fusion: Linear and Gated. They separate the effect of simple dimensional reduction from the effect of controlling how much communication information is added.

Linear fusion compresses the local state from 256 dimensions to 64 dimensions without applying ReLU, and then combines this result directly with the received communication information. It is used to test whether dimensional reduction alone is sufficient.

Gated fusion learns a set of weights between zero and one that act like adjustable switches. Based on the local state and the received information, these switches decide how much of each part of the communication information should be added to the local state. It is used to test whether controlling the amount of incoming information is more effective than direct fusion.

### 3.3 Grouping Method Settings

As explained in Chapter 2, this task has no correct actor-group labels, so it requires unsupervised grouping. The grouping-method ablation compares K-means as the baseline with three alternatives: GMM, agglomerative clustering, and balanced spectral clustering. GMM was included to allow groups with different spreads and some overlap; agglomerative clustering was included to remove random initialization; and balanced spectral clustering was included to test whether directly balancing group size could improve the result.

The grouping-method ablation used one common 5F configuration to compare K-means, GMM, agglomerative clustering, and balanced spectral clustering. All four methods used exactly the same feature input, TarMAC Hybrid fusion, random seed, grouped MAPPO training



procedure, and evaluation procedure. This controlled design keeps the grouping method as the main difference between the four experiments.

The grouping-method ablation uses the feature matrix  $X \in \mathbb{R}^{25 \times 5}$ . Each row represents one building, and the five columns are battery capacity, total HVAC rated power, mean heating demand, mean non-shiftable load, and mean lower-comfort-bound exceedance. The same combination is supplied to all four methods for a fair comparison. In the feature ablations, the matrix has the more general form  $X \in \mathbb{R}^{25 \times d}$ , where  $d$  changes with the selected feature columns. Once a feature combination is fixed for one comparison, every run uses exactly the same data-derived matrix; the matrix values are not randomly generated.

**K-means.** For both four and five candidate groups, K-means assigns each building to its nearest centre and updates the centres repeatedly. Each fit tries ten initial centre settings to reduce the effect of an unsuitable random start. The best balanced candidate is retained.

**GMM.** GMM estimates the probability that each building belongs to each group and assigns it to the group with the highest probability. It is tested with both four and five groups. Each fit uses three starting points and allows each group to have its own width and direction; this is the purpose of the full covariance matrix. A small  $10^{-6}$  adjustment is added to prevent unstable numerical calculations.

**Agglomerative clustering.** This method starts with each building in its own group and repeatedly merges the two groups that are most similar. Ward’s rule selects the merge that keeps the buildings inside the new group as similar as possible. The process stops when four or five groups remain. It does not use a random starting point, so the same feature matrix produces the same result.

**Balanced spectral clustering.** This method first gives a stronger connection to buildings with more similar features, and then uses the connection pattern to form groups. The Hungarian assignment used by balanced spectral reserves a fixed number of places for each group. It is simply a matching method: it considers the possible building-to-place assignments and finds the one with the smallest total assignment difference. This assigns every building while keeping group sizes equal or different by at most one. If two candidates are equally balanced, the lower normalized-cut cost is preferred because it means that strongly connected buildings are kept together more successfully.

Before any method is run, every selected feature column is standardized. The standardization and the basic group-balance score are

$$X_{ij} = \frac{f_{ij} - \mu_j}{\sigma_j}, \quad (3.11)$$

$$S_{\text{balance}} = \frac{\min_k n_k}{\max_k n_k}, \quad (3.12)$$

where  $f_{ij}$  is the original value of feature  $j$  for building  $i$ , and  $\mu_j$  and  $\sigma_j$  are the mean and standard deviation of that feature over the 25 buildings. Standardization stops a feature with large numerical units from dominating the grouping. In the balance score,  $n_k$  is the number of buildings in group  $k$ ; a value closer to one means that the group sizes are more similar. The experiments treat candidate groupings containing an empty group or a group with only one building separately. If a candidate contains an empty group, its selection score is set directly to  $-1$ ; if its smallest group contains only one building,  $0.5$  is subtracted from the ratio above. The selection score is a value used to compare whether buildings are distributed evenly across the different candidate groupings. Normally, it is the balance score above, namely the smallest group size divided by the largest group size. Finally, the candidate with the highest selection score is retained.

### 3.4 Grouping Feature Design

Based on the results of the preceding grouping-method experiment, agglomerative clustering performed well and was fixed as the grouping method for the subsequent feature experiments. This keeps the clustering algorithm unchanged while the selected feature columns are varied.

The initial experiment considered 37 possible features: seven static building features and 30 operational features. The static features describe battery and HVAC size, photovoltaic system size and roof area, and whether different thermal storage systems are installed. The operational features describe non-shiftable load, cooling, heating, domestic hot water, solar generation, occupancy, indoor temperature, and comfort-bound exceedance using statistics such as the mean, standard deviation, maximum, and sum.

Before the compact ablations, these 37 features were screened to nine. Three principles were used. First, the retained features should be directly related to the available control flexibility,

electrical demand, thermal demand, or indoor comfort. Second, highly repetitive descriptions of the same signal were reduced: mean and maximum values were kept to represent typical and peak demand, while several standard-deviation and sum features were removed. Third, the selection matched the Vermont winter experiment, so heating demand and lower-comfort exceedance were more informative than cooling demand and upper-comfort exceedance. PV, solar, storage-presence flags, and other secondary descriptors were not included in the compact pool because they were less direct for this specific control question or repeated information already represented by the retained demand and capacity variables. This screening produced the following nine control-profile features:

- `bes_capacity_kwh`: battery energy capacity, representing how much electrical energy the building can shift over time;
- `hvac_total_kw`: total rated HVAC power, representing the installed ability to change heating or cooling electricity use;
- `heating_mean`: average heating demand, representing the building’s typical thermal load;
- `heating_max`: maximum heating demand, representing its peak thermal load;
- `ns1_mean`: average non-shiftable load, representing typical background electricity demand;
- `ns1_max`: maximum non-shiftable load, representing peak background demand;
- `occupants_mean`: average occupancy, representing typical building use intensity;
- `indoor_temp_std`: standard deviation of indoor temperature, representing how much indoor temperature varies;
- `comfort_lower_excess_mean`: average amount by which indoor temperature falls below the lower comfort limit, representing under-heating behaviour.

The full 9F control-profile description consists of all nine variables above. The experiment first used all nine features in an agglomerative-clustering grouped TarMAC Hybrid MAPPO experiment. The input features were then reduced progressively:

- 5F: battery capacity, HVAC rated power, mean heating demand, mean non-shiftable load, and mean lower-comfort-bound exceedance;
- 4F: the first four features above, with the comfort feature removed;
- 3F: battery capacity, mean heating demand, and mean non-shiftable load, with HVAC rated power also removed.

This reduction tests whether adding more building descriptions provides useful grouping information or introduces redundant information.

After the feature-count ablation, the experimental results showed that the 3F settings generally performed better. A detailed analysis is provided in Chapter ??.

Choosing any three variables from the nine-feature pool gives  $\binom{9}{3} = 84$  possible combinations. Training every combination with several random seeds would be expensive, so nine representative combinations were selected using physical reasoning and controlled substitutions:

- A: capacity + heating mean + NSL mean, covering storage flexibility, typical thermal demand, and typical background load;
- B: capacity + HVAC rating + NSL mean, testing installed HVAC capability instead of measured heating demand;
- C: capacity + heating maximum + NSL mean, testing peak rather than average heating demand;
- D: capacity + heating mean + NSL maximum, testing peak rather than average non-shiftable load;
- E: capacity + HVAC rating + NSL maximum, combining installed HVAC capability with peak background demand;
- F: capacity + heating maximum + NSL maximum, combining the two peak-demand descriptions;
- G: capacity + heating mean + lower-comfort exceedance, testing whether a direct comfort-related feature is more useful than non-shiftable load;

- H: capacity + heating mean + indoor-temperature variation, testing whether indoor thermal stability helps to define the groups;
- I: heating mean + NSL mean + lower-comfort exceedance, removing capacity to test whether battery size is essential to the grouping.

These combinations cover the main questions of installed capability versus observed demand, mean versus peak demand, electrical load versus comfort and temperature information, and grouping with or without battery capacity. Average occupancy was not included in the nine compact combinations because its effect is partly reflected in heating and non-shiftable load, while those two variables are more directly related to the control task. The selected combinations are therefore a focused screening of the most plausible choices rather than an exhaustive search of all 84 possibilities.

For Texas, the same feature roles are retained but `heating_mean` is replaced with `cooling_mean`. Thus the cross-climate experiment tests a transferable principle—capacity, controllable thermal burden, and background load—rather than blindly copying a Vermont heating feature into a cooling-dominated season.

### 3.5 Soft Router for Dynamic Actor Selection

Fixed grouping assigns each building to one actor before training, and that actor does not change when the building’s battery level, load, or heating demand changes. Its performance therefore depends strongly on the quality of the initial grouping. Soft Router was introduced to address this limitation. Its purpose is to let each building choose the most suitable actor for its current condition, while still making use of the group actors that have already been trained. It therefore tests whether actor selection can become more flexible and less dependent on one fixed grouping.

The Soft Router itself is a small shared neural network: for each building, it reads the building’s current information and outputs one selection probability for every candidate actor. Each trained group actor is one such candidate, also called an expert. To explain how the router chooses among them, first consider how one expert produces an action. At every time step, a building provides an observation  $o_{it}$ , such as its battery SOC, current load, heating demand,

indoor temperature, and time information. The expert’s *encoder* turns these raw values into a short internal summary. TarMAC adds useful information from the other buildings, and the *action head* finally converts the communicated summary into a control action.

The encoder and action head of each pretrained expert were learned together. Some early routers switched only the action head, so a head could receive an internal summary produced by an encoder with which it had never been trained. The stable version instead switches the matched encoder and action head together; this is called *full-expert routing*. It preserves the complete control behaviour of each pretrained actor and is therefore easier to keep stable. The following explanation focuses on this version, while the other router designs are summarized later.

To choose a full expert, the router receives an input with two complementary parts. First, the encoder of the actor originally assigned to each building converts its observation into a 256-dimensional latent feature. TarMAC then communicates among all 25 buildings. Thus, the communicated feature for building  $i$  can be written as

$$h_{it} = \text{TarMAC}_i\left(\{\text{Enc}_{c_j}(o_{jt})\}_{j=1}^N\right) \in \mathbb{R}^{256}, \quad (3.13)$$

where  $c_j$  is the original fixed group of building  $j$ . Although  $h_{it}$  is associated with building  $i$ , it contains information received from the other buildings. It is a learned summary rather than a directly measured physical quantity.

The learned summary is rich, but an encoder trained mainly for action generation may compress or obscure physical quantities that are especially important for choosing between experts. The capacity-aware router therefore appends seven explicit features for the same building  $i$ ,

$$q_{it} = [\bar{C}_i, \bar{H}_i, s_{it}, \bar{C}_i(1 - s_{it}), \bar{C}_i s_{it}, \ell_{it}, u_{it}] \in \mathbb{R}^7, \quad (3.14)$$

where  $\bar{C}_i$  and  $\bar{H}_i$  are its normalized battery capacity and HVAC rating,  $s_{it}$  is its SOC,  $\ell_{it}$  is its current non-shiftable load, and  $u_{it}$  is its current heating demand. The fourth and fifth entries explicitly represent remaining charging space and available discharge energy. These seven values belong only to building  $i$ ; they are not a concatenation of the seven features of all buildings. Cross-building information is already supplied by  $h_{it}$ .

The complete router input is therefore

$$x_{it}^{\text{router}} = [h_{it}; q_{it}] \in \mathbb{R}^{263}. \quad (3.15)$$

There is one shared router network, not one separately parameterized router per building. The same parameters  $\theta$  are applied to all 25 inputs, while the different building states produce 25 different routing distributions. In the reported stable full-expert run, this feed-forward network has the form  $263 \rightarrow 64 \rightarrow 5$ : a linear layer, ReLU, and a second linear layer that produces one logit for each of the five experts. For building  $i$  at time  $t$ , the logits are converted into probabilities,

$$d_{it} = \text{softmax}(\text{Router}_{\theta}(x_{it}^{\text{router}})/\tau), \quad (3.16)$$

where every element of  $d_{it}$  is the probability of selecting one actor. The probabilities add up to one. The temperature  $\tau$  controls how decisive the choice is: a lower value makes the router concentrate more strongly on one actor, whereas a higher value spreads the probabilities more evenly.

At the beginning of training, immediately abandoning the original grouping can destroy the useful behaviour already learned by the actors. The router therefore combines its new probabilities with the original fixed assignment,

$$g_{it} = (1 - \alpha)p_i + \alpha d_{it}. \quad (3.17)$$

Here,  $p_i$  marks the actor originally assigned to building  $i$ , and  $\alpha$  controls how much freedom is given to the router. When  $\alpha = 0$ , the original fixed actor is used. As  $\alpha$  increases, the router is allowed to choose other actors more often. The experiments pass  $\tau$ , the starting and final values of  $\alpha$ , the warm-up length, the router learning rate, and the balance strength as training settings. This makes it possible to change how quickly and how strongly dynamic selection is introduced without changing the building data.

During training, an expert is selected according to  $g_{it}$  and generates the building action. To protect the useful behaviour in the pretrained actors, the stable experiment uses staged training: it first freezes all experts and updates only the router. This raises an important question: if the experts no longer change, how can a policy loss still train the router? The reason is that the complete policy includes both the expert action distributions and the router’s selection probabilities. Let  $\pi_{\phi_k}(a_{it} \mid s_t)$  denote the action distribution of expert  $k$ , with expert

parameters  $\phi_k$ . The complete routed policy is the mixture

$$\pi_{\theta,\phi}(a_{it} \mid s_t) = \sum_{k=1}^K g_{itk}(\theta) \pi_{\phi_k}(a_{it} \mid s_t). \quad (3.18)$$

Freezing an expert fixes  $\phi_k$  and its action distribution, but it does not fix the complete policy. Changing the router parameters  $\theta$  changes the mixture weights  $g_{itk}$  and therefore changes the probability of the executed action. Equation (3.18) is therefore the link between staged freezing and the PPO loss derived below.

CityLearn executes the actions and returns the same reward defined in Equation (3.5); the router has no separate reward. The shared part measures portfolio load tracking, and the building-specific part measures indoor comfort. After one rollout, MAPPO uses the rewards and the critic to form an advantage estimate, which can be summarized as

$$\hat{A}_{it} = \hat{R}_{it} - V_{\psi}(s_t). \quad (3.19)$$

A positive advantage means that the sampled action led to a better result than the critic expected, while a negative advantage means that it was worse. For the stored action, PPO recomputes the routed-policy probability and forms the ratio

$$r_{it}(\theta, \phi) = \frac{\pi_{\theta,\phi}(a_{it} \mid s_t)}{\pi_{\text{old}}(a_{it} \mid s_t)}. \quad (3.20)$$

The clipped policy loss is

$$\mathcal{L}_{\text{PPO}} = -\mathbb{E}_{i,t} \left[ \min \left( r_{it} \hat{A}_{it}, \text{clip}(r_{it}, 1 - \epsilon, 1 + \epsilon) \hat{A}_{it} \right) \right]. \quad (3.21)$$

Even when  $\phi_k$  is frozen, Equation (3.18) shows that  $r_{it}$  and the PPO loss still depend on  $\theta$ . The gradient path is

$$\frac{\partial \mathcal{L}_{\text{PPO}}}{\partial \theta} = \frac{\partial \mathcal{L}_{\text{PPO}}}{\partial \pi} \frac{\partial \pi}{\partial g} \frac{\partial g}{\partial \theta}. \quad (3.22)$$

Thus, the reward itself is not differentiated through CityLearn. It determines  $\hat{A}_{it}$ , which tells PPO whether to increase or decrease the probability of the stored action; automatic differentiation then converts this probability error into a gradient for the router.

For example, suppose two frozen experts assign probabilities 0.8 and 0.2 to the executed action, and the router initially gives them weights 0.5 and 0.5. The mixture assigns probability



$0.5(0.8) + 0.5(0.2) = 0.5$  to that action. If its advantage is positive, increasing the first router weight to 0.6 and decreasing the second to 0.4 raises the mixture probability to  $0.6(0.8) + 0.4(0.2) = 0.56$ , without changing either expert. This is the source of the policy loss and router gradient during the frozen-expert stage.

The implemented actor objective also includes exploration and routing-balance terms,

$$\mathcal{L}_{\text{actor}} = \mathcal{L}_{\text{PPO}} - c_H \mathcal{H}(\pi) + c_B \mathcal{L}_{\text{balance}}, \quad (3.23)$$

$$\mathcal{L}_{\text{balance}} = \frac{1}{K} \sum_{k=1}^K (\bar{g}_k - \bar{p}_k)^2, \quad (3.24)$$

where  $\bar{g}_k$  is mean expert usage in the batch and  $\bar{p}_k$  is the original group proportion. The balance term discourages routing collapse but is not an additional CityLearn reward. The critic remains trainable and continues to improve the value estimate even while the expert actors are frozen.

Several safeguards work together to reduce routing collapse. The router starts with nearly equal expert probabilities, and training samples experts from these probabilities instead of always taking the largest one. The entropy term discourages the probabilities from becoming too concentrated, the original-group prior and gradual increase of  $\alpha$  preserve early diversity, the balance loss discourages one expert from receiving nearly all buildings, and PPO clipping limits abrupt probability changes. These measures reduce, but do not completely eliminate, the possibility that an expert is rarely selected.

Freezing improves stability because it removes a moving-target problem. If the router and every expert change rapidly at the same time, the router is trying to learn which expert is suitable while the meaning and behaviour of every expert are also changing. With pretrained experts fixed, their action distributions are reliable reference points, and the smaller router only has to learn how to combine them. Together with the safeguards above, this staged freezing makes learning more stable. In the stable full-expert experiment, episodes 1–500 update the router while freezing all experts; episodes 501–1000 freeze the learned router and update only the small action heads at a low learning rate. Keeping the encoders and TarMAC module fixed preserves their pretrained latent spaces and communication behaviour. During deterministic testing, each building directly uses the expert with the largest routing probability.

## 3.6 Router Training Sequence

The soft-router experiments evolved through four designs:

1. **End-to-end legacy training.** Router and actors were trained together from randomly initialized policies. Temperature, warm-up duration, and final prior weight were varied. These runs established that a router could assign different selection probabilities to the experts, but policy learning was unstable.
2. **Pretrained two-stage training.** A fixed-group checkpoint using the strong 3fA feature combination identified earlier was loaded first. Experts were frozen for 200 episodes before limited joint fine-tuning, with temperature 0.5 or 0.7, final dynamic weight 1.0 or 0.7, and with/without the seven capacity-aware router inputs.
3. **Three-stage training.** Experiments separated static actor training, router-only learning, and dynamic actor adaptation. Both a shared- encoder version and a full-expert version were tested. The shared version attempted to put every head in one latent space; the full-expert version retained matched encoder–communication–head branches.
4. **Stable two-stage enhancement.** The best fixed 3fA checkpoint was treated as an anchor. The most conservative run trained only the router for 500 episodes (`routeronly500`). The stronger stable-head family then allowed only action heads to adapt at  $10^{-5}$  while encoders and TarMAC remained frozen and the router was frozen during the adaptation phase.

This sequence tests the central question directly. If soft routing could replace clustering, the end-to-end and shared three-stage runs should learn good experts without a strong prior. If it is instead an enhancement, the pretrained and stable variants should be better because they preserve a useful initial partition.

## 3.7 Completed Experiments

At the current stage, the following experiments have been completed:

- reward-weight comparisons using the baseline, stricter-comfort, controlled degree-hours, and load-tracking-focused parameter settings;
- comparison of the original TarMAC communication with TarMAC Hybrid, as well as the ReLU, Linear, and Gated fusion ablations;
- grouping-method experiments comparing K-means, GMM, agglomerative clustering, and balanced spectral clustering under the same 5F setting;
- grouping-feature experiments covering the full 9F control profile, the 5F, 4F, and 3F feature-count ablations, and the nine representative 3F combinations A–I with multiple random seeds;
- multi-seed stability and cross-climate experiments for the selected 3fA feature roles in Vermont and Texas;
- Soft Router experiments that gradually improved the training process from an unstable initial design to the final stable version. The final version first keeps the pretrained actors unchanged while training the selector, and then keeps the learned selector fixed while making small adjustments to the actor output layers.

## 3.8 Current Limitations

The final stable Soft Router can be trained reliably and provides a useful performance enhancement over fixed grouping. Its main limitation is that this stability depends on strong pretrained actors produced by an earlier fixed grouping; it still cannot learn equally strong actors and routing from random initialization. Therefore, Soft Router currently improves an existing grouping rather than replacing clustering completely. Validation on more climates and portfolio sizes is still needed.

# Chapter 4

## Evaluation and Result Analysis

This chapter evaluates the methods introduced in Chapter 3 using the CityLearn Annex 96 Common Exercise 1 portfolio load-tracking task. It compares reward-weight settings, TarMAC Hybrid, alternative building-grouping methods, grouping-feature ablations, Vermont–Texas generalization, and the Soft Router. The results are analysed in terms of load tracking, thermal comfort, and stability across multiple random seeds.

### 4.1 Evaluation Scope and Metrics

Vermont uses January for training/group-feature extraction and February for testing. Texas uses August for training/group features and September for testing. Unless otherwise stated, runs use 500 training episodes. Load tracking is reported using CV-RMSE and the absolute value of NMBE. Comfort is reported as both the percentage of building-hours outside the seasonal comfort band and degree-hours per building-day, which preserves the magnitude as well as the duration of a temperature violation while making results comparable across test periods and portfolio sizes. It is calculated as

$$\text{DH}_{\text{bd}} = \frac{\text{portfolio total degree-hours}}{N_{\text{building}} D_{\text{test}}} = \frac{\sum_{i=1}^{N_{\text{building}}} \sum_{t=1}^T d_{i,t} \Delta t}{N_{\text{building}} D_{\text{test}}}, \quad (4.1)$$

where  $d_{i,t}$  is the number of degrees by which building  $i$  is outside the comfort band at time  $t$ ,  $\Delta t$  is the time-step duration,  $N_{\text{building}}$  is the number of buildings, and  $D_{\text{test}}$  is the number of test days. Its unit is °C h per building-day. Lower values are better for all four metrics.

The evaluation combines repeated training results from multiple random seeds across the main feature, climate, and Soft Router studies. Controlled ablations compare runs under the same seed and settings, while repeated runs are summarized by their mean and standard deviation to verify that the main conclusions do not depend on one unusually good result. An individual run is discussed only when it is needed to explain a specific controlled comparison.

## 4.2 TarMAC Hybrid Improvement

Table 4.1 compares the original TarMAC controller and the three Hybrid fusion variants under the previous seed-42 benchmark. The ReLU Hybrid reduces CV-RMSE from 49.65% to 47.01%, reduces comfort exceedance from 22.10% to 17.99%, and improves reward from  $-5996$  to  $-4715$ . Relative to the original TarMAC values, these are a 5.3% reduction in CV-RMSE, an 18.6% reduction in comfort exceedance, and a 21.4% reduction in reward magnitude. Absolute NMBE changes only slightly, from 3.32% to 3.51%.

| Communication   | Primary rank | CV-RMSE<br>(%) | NMBE <br>(%) | Comfort<br>exceed. (%) | Reward<br>sum |
|-----------------|--------------|----------------|--------------|------------------------|---------------|
| Original TarMAC | 63           | 49.65          | 3.32         | 22.10                  | $-5996$       |
| Hybrid, ReLU    | <b>29</b>    | <b>47.01</b>   | 3.51         | <b>17.99</b>           | $-4715$       |
| Hybrid, linear  | 74           | 50.39          | 3.13         | 22.20                  | $-6000$       |
| Hybrid, gated   | 69           | 48.74          | <b>3.06</b>  | 25.35                  | $-6923$       |

Table 4.1: Original TarMAC and TarMAC Hybrid fusion ablation, Vermont seed 42.

The linear and gated controls show that an additional layer alone is not enough. The strongest initial Hybrid result specifically combines the compact local projection, ReLU, and residual update. Later grouping experiments use the linear Hybrid mode as a controlled communication backbone, as indicated by `linear` in their run names; the term “Hybrid” in those experiments therefore refers to the local/context residual architecture family, not to an assumption that every run uses ReLU.

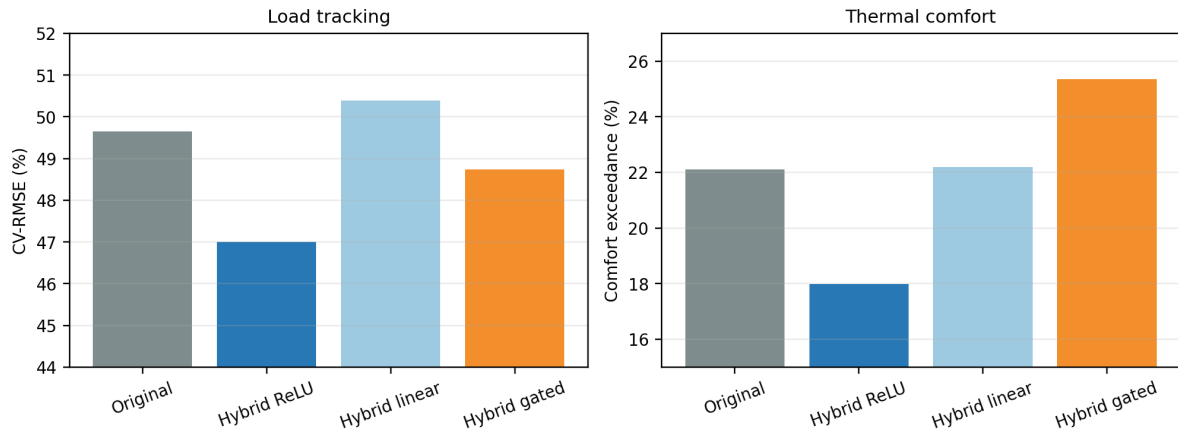


Figure 4.1: TarMAC Hybrid ablation using the consolidated experiment metrics.

### 4.3 Comparison of Grouping Methods

Table 4.2 compares the four grouping methods using the same 5F feature set and linear Hybrid backbone. Each method is summarized over the same three seeds (0, 1, and 42), allowing the grouping methods to be compared under the same experimental conditions.

| Method            | CV-RMSE<br>(%) | NMBE <br>(%) | Degree-hours<br>(°C h/building-day) | Comfort<br>exceed. (%) | Reward<br>sum |
|-------------------|----------------|--------------|-------------------------------------|------------------------|---------------|
| K-means           | 50.90          | <b>1.36</b>  | <b>3.64</b>                         | <b>15.86</b>           | −4326         |
| GMM               | <b>48.11</b>   | 1.67         | 5.23                                | 21.38                  | −5583         |
| Agglomerative     | 49.85          | 1.62         | 5.42                                | 20.96                  | −5574         |
| Balanced spectral | 51.23          | 4.22         | 11.50                               | 25.72                  | −7496         |

Table 4.2: Three-seed mean comparison of the selected fixed grouping methods with 5F and linear Hybrid, using matched seeds 0, 1, and 42.

The mean results show that GMM has the lowest CV-RMSE (48.11%), while K-means has the lowest absolute NMBE (1.36%), degree-hours per building-day (3.64), and comfort exceedance (15.86%), as well as the least negative reward (−4326). Agglomerative clustering remains competitive, with a CV-RMSE of 49.85% and an absolute NMBE of 1.62%, whereas balanced spectral gives the weakest comfort and bias results in this comparison. No single

method is best on every metric. Agglomerative clustering provides competitive load-tracking performance and is therefore used as the fixed grouping method for the later feature ablations, ensuring that those experiments compare only the effect of changing the feature set.

## 4.4 Feature-Count Ablation

Table 4.3 compares 3fA, 4F, and 5F with agglomerative grouping and the linear-Hybrid backbone. The 3fA and 5F results are five-seed means, while the 4F result is evaluated over four independent seeds. The 3fA mean is best on all four reported metrics. The 5F mean has a similar CV-RMSE to 3fA but larger bias and comfort errors, and the reported 4F mean is also worse than the 3fA mean on every metric. Overall, the results support the low-redundancy hypothesis: even when each feature is individually relevant, adding more standardized features can still change the grouping result and reduce performance.

Figure 4.2 summarizes all four primary metrics rather than displaying CV-RMSE alone. For each lower-is-better metric, a candidate’s relative score is the best mean value in the panel divided by that candidate’s mean value, multiplied by 100. The four relative scores for CV-RMSE, absolute NMBE, degree-hours, and comfort exceedance are then averaged with equal weight:

$$S_j = \frac{100}{4} \sum_{m=1}^4 \frac{\min_k \bar{x}_{k,m}}{\bar{x}_{j,m}}. \quad (4.2)$$

Higher scores therefore indicate a better overall balance. In the feature-count comparison, 3fA ranks first with a score of 100.0, ahead of 4F at 80.5 and 5F at 75.4. The score is used only to summarize the four reported metrics visually; the raw values remain available in Table 4.3.

| Set | Added information             | CV-RMSE<br>(%) | NMBE <br>(%) | Degree-hours<br>(°C h/building-day) | Comfort<br>exceed. (%) |
|-----|-------------------------------|----------------|--------------|-------------------------------------|------------------------|
| 3fA | capacity, heat mean, NSL mean | <b>49.86</b>   | <b>0.81</b>  | <b>4.64</b>                         | <b>19.21</b>           |
| 4F  | 3fA + HVAC rating             | 51.62          | 1.80         | 5.27                                | 20.82                  |
| 5F  | 4F + comfort lower excess     | 49.97          | 2.01         | 6.36                                | 21.74                  |

Table 4.3: Feature-count ablation with agglomerative grouping. The 3fA and 5F rows are five-seed means, while the 4F row is a four-seed mean.

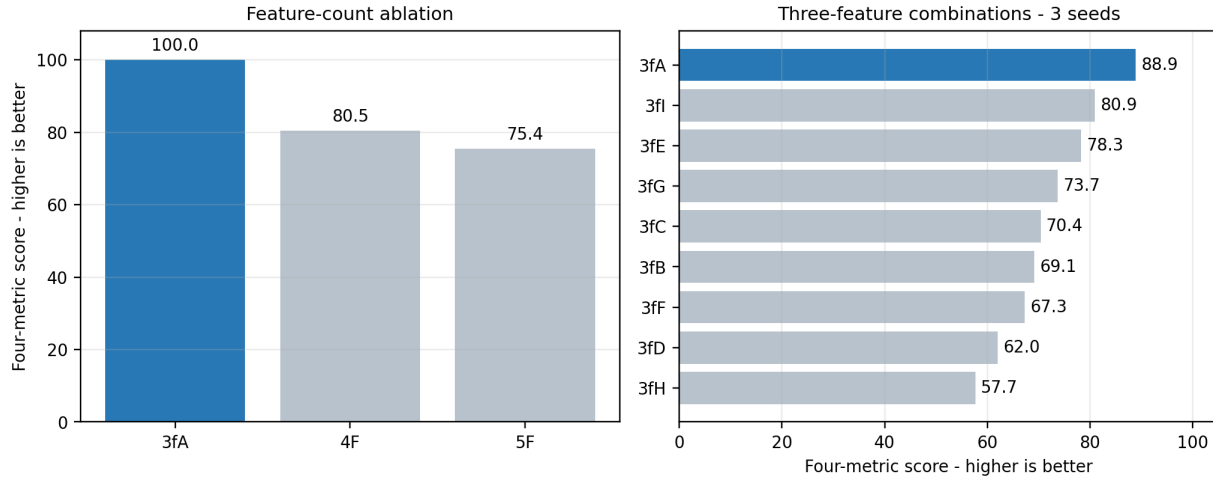


Figure 4.2: Equal-weight four-metric scores for the feature-count and three-feature-combination ablations. Higher is better, and 3fA is highlighted in blue. Scores are calculated from the recorded CV-RMSE, absolute NMBE, degree-hours, and comfort-exceedance results using Equation (4.2). The feature-count panel uses five runs for 3fA and 5F and four runs for 4F.



## 4.5 Three-Feature Combination Ablation

Table 4.4 reports mean and sample standard deviation over seeds 0–2. The experiment is multi-objective, so “best” depends on the metric. 3fI, which removes capacity and adds comfort, has the lowest mean CV-RMSE and degree-hours. 3fE has the lowest comfort variability. In contrast, 3fA has by far the lowest and most stable absolute bias,  $0.49 \pm 0.18\%$ , and its features remain directly interpretable across climates. It is therefore selected as the best overall compromise.

The right panel of Figure 4.2 confirms this overall selection using the same four-metric score. Across the nine three-feature combinations, 3fA ranks first at 88.9, followed by 3fI at 80.9 and 3fE at 78.3. This does not mean that 3fA is best on every individual metric; rather, its much lower load bias compensates for the individual CV-RMSE and comfort advantages of some alternatives and gives it the strongest equal-weight overall result.

| Set | Three features (capacity is included unless stated) | CV-RMSE<br>(%)                     | NMBE <br>(%)                      | Degree-hours<br>(°C h/building-day) | Comfort<br>exceed. (%) |
|-----|-----------------------------------------------------|------------------------------------|-----------------------------------|-------------------------------------|------------------------|
| A   | capacity, heating mean, NSL mean                    | $51.93 \pm 2.46$                   | <b><math>0.49 \pm 0.18</math></b> | $5.22 \pm 1.21$                     | $20.27 \pm 2.44$       |
| B   | capacity, HVAC, NSL mean                            | $52.59 \pm 4.50$                   | $1.00 \pm 0.98$                   | $7.51 \pm 2.50$                     | $21.63 \pm 2.48$       |
| C   | capacity, heating max, NSL mean                     | $49.39 \pm 3.99$                   | $1.52 \pm 1.26$                   | $7.09 \pm 2.09$                     | $18.61 \pm 3.74$       |
| D   | capacity, heating mean, NSL max                     | $53.57 \pm 3.29$                   | $2.01 \pm 3.02$                   | $8.02 \pm 4.90$                     | $21.18 \pm 3.03$       |
| E   | capacity, HVAC, NSL max                             | $53.24 \pm 2.45$                   | $1.38 \pm 0.86$                   | $4.20 \pm 0.98$                     | $18.88 \pm 1.53$       |
| F   | capacity, heating max, NSL max                      | $49.64 \pm 2.58$                   | $2.48 \pm 1.07$                   | $6.07 \pm 1.42$                     | $20.43 \pm 2.59$       |
| G   | capacity, heating mean, comfort                     | $49.47 \pm 1.70$                   | $1.52 \pm 0.56$                   | $5.33 \pm 1.88$                     | $19.49 \pm 3.01$       |
| H   | capacity, heating mean, temp. std                   | $52.09 \pm 4.08$                   | $2.96 \pm 2.69$                   | $9.70 \pm 6.29$                     | $22.24 \pm 4.59$       |
| I   | heating mean, NSL mean, comfort (no capacity)       | <b><math>48.93 \pm 2.31</math></b> | $2.09 \pm 1.01$                   | <b><math>3.77 \pm 1.42</math></b>   | $18.06 \pm 4.86$       |

Table 4.4: Three-feature combination ablation, agglomerative grouping, seeds 0–2.

The larger five-run 3fA stability set (seeds 0, 1, 2, 3, and 42) gives CV-RMSE  $49.86 \pm 3.65\%$ , |NMBE|  $0.81 \pm 0.68\%$ , degree-hours  $4.64 \pm 1.27$  per building-day, comfort exceedance  $19.21 \pm 2.85\%$ , and reward  $-5125 \pm 731$ . Its CV-RMSE range is 44.62–54.09%, while all five absolute NMBE values remain at or below 2.01%. This supports stability in portfolio bias, although seed sensitivity in load shape has not disappeared.

## 4.6 Vermont–Texas Generalization

The cross-climate experiment transfers the 3fA *roles*, not the trained Vermont parameters. Vermont uses capacity–heating mean–NSL mean; Texas uses capacity–cooling mean–NSL mean, with its own August training and September test. Table 4.5 gives the matched three-seed runs with the comfort15/binary30/degree10 reward setting.

| Climate | CV-RMSE<br>(%)   | NMBE <br>(%)     | DH/building-day<br>(°C h) | Comfort exceed.<br>(%) |
|---------|------------------|------------------|---------------------------|------------------------|
| VT      | $49.64 \pm 1.59$ | $3.04 \pm 1.49$  | $5.10 \pm 1.95$           | $18.74 \pm 1.90$       |
| TX      | $78.83 \pm 4.90$ | $45.57 \pm 1.98$ | $1.59 \pm 0.31$           | $11.22 \pm 1.66$       |

Table 4.5: 3fA role-based generalization; mean  $\pm$  sample standard deviation over three seeds. VT uses capacity–heating mean–NSL mean, and TX uses capacity–cooling mean–NSL mean.

The numerical scale differs because the two climates and target profiles are different, so VT and TX should not be ranked directly against each other. The important result is stability within each climate and improvement over the non-learning references. In TX, the three CV-RMSE values are 73.78–83.57%, compared with 163.74% for the no-control baseline and 207.63% for RBC. Mean absolute NMBE is 45.57%, compared with 79.12% and 89.57%, respectively. TX bias therefore remains a clear limitation, but the learned controller is still substantially better than both references and has low seed variation in comfort and bias. This supports generalization of the compact feature principle.

## 4.7 Soft-Router Ablation and Training Progression

Table 4.6 lists the tested Soft Router variants, and Figure 4.3 shows the performance trend across the main development stages. Single-run rows use seed 42; the final stable-head row is the mean and sample standard deviation over three router-training seeds. The intermediate variants were evaluated only with seed 42 because their results were already poor and repeating these long training runs would require substantial time and computational cost. Multi-seed evaluation was therefore reserved for the final stable version.

| Stage / variant         | Main change                                                 | CV-RMSE<br>(%)                     | NMBE <br>(%)    | Degree-hours<br>(°C h/building-day) | Comfort<br>exceed. (%)             |
|-------------------------|-------------------------------------------------------------|------------------------------------|-----------------|-------------------------------------|------------------------------------|
| Legacy end-to-end       | $\tau = 0.5$ , warm-up 50                                   | 47.60                              | 4.31            | 10.02                               | 24.64                              |
| Two-stage freeze200     | pretrained experts, $\tau = 0.5$ , $\alpha \rightarrow 1$   | 48.26                              | 1.35            | 6.15                                | 21.76                              |
| Two-stage prior0.7      | pretrained experts, $\tau = 0.7$ , $\alpha \rightarrow 0.7$ | 46.54                              | 1.02            | 6.60                                | 22.05                              |
| Two-stage no capacity   | remove seven physical router inputs                         | 46.94                              | <b>0.57</b>     | 5.53                                | 19.54                              |
| Three-stage shared      | shared encoder, 1500 episodes                               | 52.44                              | 7.48            | 9.27                                | 31.35                              |
| Three-stage full expert | preserve complete expert branches                           | 45.45                              | 2.28            | 6.39                                | 20.42                              |
| Stable router-only      | freeze experts for all 500 episodes                         | 45.99                              | 1.22            | 4.16                                | 19.17                              |
| Stable heads, 3 seeds   | router 500 + head-only adaptation                           | <b>45.12 <math>\pm</math> 0.35</b> | 1.35 $\pm$ 0.33 | <b>3.78 <math>\pm</math> 0.44</b>   | <b>17.30 <math>\pm</math> 1.33</b> |

Table 4.6: Soft-router development and ablation. Single rows use seed 42 unless stated.

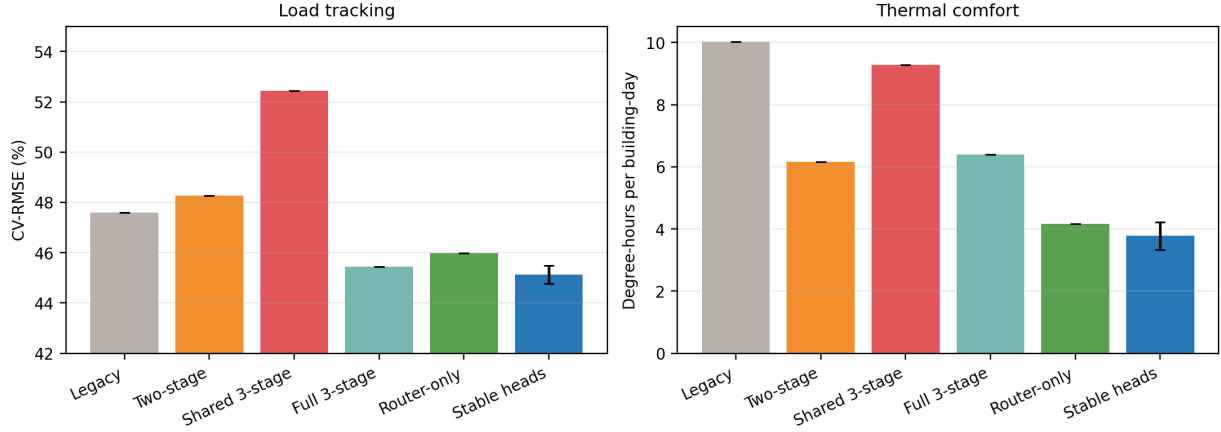


Figure 4.3: Primary metrics across the main soft-router development stages.

### 4.7.1 Results for Each Soft Router Variant

The legacy end-to-end version trains the router and actors together from the beginning. Its CV-RMSE is 47.60%, but its absolute NMBE is 4.31%, its degree-hours are 10.02 per building-day, and its comfort exceedance is 24.64%. It can learn a reasonable load shape, but its load bias and thermal comfort are poor.

The two-stage versions first load the trained 3fA experts and then train the router. This immediately reduces the bias. The freeze200 version reaches 48.26% CV-RMSE and 1.35% absolute NMBE. Changing the temperature and prior in the prior0.7 version improves CV-RMSE to 46.54%. Removing the seven physical capacity inputs gives the lowest absolute NMBE in the table, 0.57%, and also reduces degree-hours to 5.53 per building-day. These results show that loading useful experts is more important than giving the router many additional inputs.

The three-stage experiments test how much of each expert should be preserved. The shared-encoder version is the worst variant, with 52.44% CV-RMSE, 7.48% absolute NMBE, and 31.35% comfort exceedance. In contrast, preserving each complete expert branch reduces CV-RMSE to 45.45%. The comparison shows that an expert’s encoder, communication path, and output head should remain together; rebuilding a shared internal representation can destroy the useful behaviour already learned by the actors.

The stable versions limit what can change after router training. The seed-42 router-only run reaches 45.99% CV-RMSE and 4.16 degree-hours per building-day. Across all five available router-only runs, CV-RMSE is  $49.83 \pm 3.34\%$ ,  $|\text{NMBE}|$  is  $1.19 \pm 0.60\%$ , degree-hours are  $4.55 \pm 1.25$  per building-day, and comfort exceedance is  $18.73 \pm 2.43\%$ . Allowing only small changes to the action heads after the router has stabilized gives the strongest final result:  $45.12 \pm 0.35\%$  CV-RMSE,  $1.35 \pm 0.33\%$  absolute NMBE,  $3.78 \pm 0.44$  degree-hours per building-day, and  $17.30 \pm 1.33\%$  comfort exceedance over three router-training seeds.

### 4.7.2 Router-Collapse Analysis

Routing collapse occurs when the router sends almost all decisions to one expert, leaving the other actors with little or no use. Table 4.7 uses the recorded training history of two matched seed-42 full-expert runs to examine this directly. Both use temperature  $\tau = 0.5$ , increase  $\alpha$  from

0.5 to 1.0 over 150 episodes, use entropy coefficient  $c_H = 0.05$ , balance coefficient  $c_B = 0.01$ , and train the router at  $10^{-4}$ . Their first 500 router-only episodes are identical. After this point, the joint version resumes router updates after a 100-episode pause and updates the full experts at  $3 \times 10^{-5}$ ; the stable version freezes the router for all remaining 500 episodes and updates only the action heads at  $10^{-5}$ .

| Recorded point                       | $\tau$ | $\alpha$ | $c_H$ | $c_B$ | Router status | Gate entropy | Mean largest gate | Largest expert share |
|--------------------------------------|--------|----------|-------|-------|---------------|--------------|-------------------|----------------------|
| Start, episode 1                     | 0.5    | 0.50     | 0.05  | 0.01  | training      | 1.223        | 0.602             | 30.6%                |
| End of router-only, episode 500      | 0.5    | 1.00     | 0.05  | 0.01  | training      | 0.275        | 0.891             | 36.4%                |
| Joint adaptation, episode 1000       | 0.5    | 1.00     | 0.05  | 0.01  | updated       | 0.171        | 0.934             | 51.7%                |
| Stable head adaptation, episode 1000 | 0.5    | 1.00     | 0.05  | 0.01  | frozen        | 0.276        | 0.891             | 37.0%                |

Table 4.7: Matched seed-42 routing-concentration data. Higher entropy and lower largest-gate/share values indicate less concentration.

With five experts, the largest possible gate entropy is  $\log(5) = 1.609$ . The prior and gradual  $\alpha$  schedule give initial expert shares of 21.9%, 20.1%, 30.6%, 13.3%, and 14.1%. At episode 500 the largest share is still only 36.4%. Continuing joint training then reduces entropy from 0.275 to 0.171 and raises one expert’s share to 51.7%, whereas freezing the router preserves entropy at 0.276 and limits the largest share to 37.0%. The three final stable seeds have largest shares of 37.5%, 40.5%, and 37.0%, and the matched seed-42 stable run also improves CV-RMSE from 45.45% to 44.72% and degree-hours from 6.39 to 4.27 per building-day. Together with probability sampling and PPO clipping, this supports the combined use of the prior, gradual  $\alpha$ , entropy, balance, and staged freezing. It does not produce perfect balance: expert 3 receives only 0.015–0.036% of final traffic. The safeguards therefore prevent total one-expert collapse, but one branch can still become almost unused.

### 4.7.3 Final Soft Router Assessment

Overall, the stable-head Soft Router is the strongest and most stable routing version. Its mean CV-RMSE and degree-hours are better than those of the five-run fixed 3fA family ( $49.86 \pm 3.65\%$  and  $4.64 \pm 1.27$  per building-day). All three stable-head runs start from the same seed-42 pretrained actors and only change the random seed used for the later router training. Therefore, these results show that the router can be trained consistently, but they do not test whether the

whole model is equally stable when trained from the beginning with different seeds. Therefore, the final stable-head Soft Router is a useful and stable improvement built on strong fixed experts, but it does not remove the need for good initial grouping.

## 4.8 Detailed Results of the Best-Performing Experiments

The following plots present detailed test results from the selected best-performing fixed 3fA and stable Soft Router experiments. The fixed 3fA seed-42 run is shown because its checkpoint supplies the pretrained actors used by the final router experiments. Stable Soft Router seed 0 is shown because it has the best primary rank among the three final stable-head runs. These plots show individual experiments rather than curves averaged across seeds.

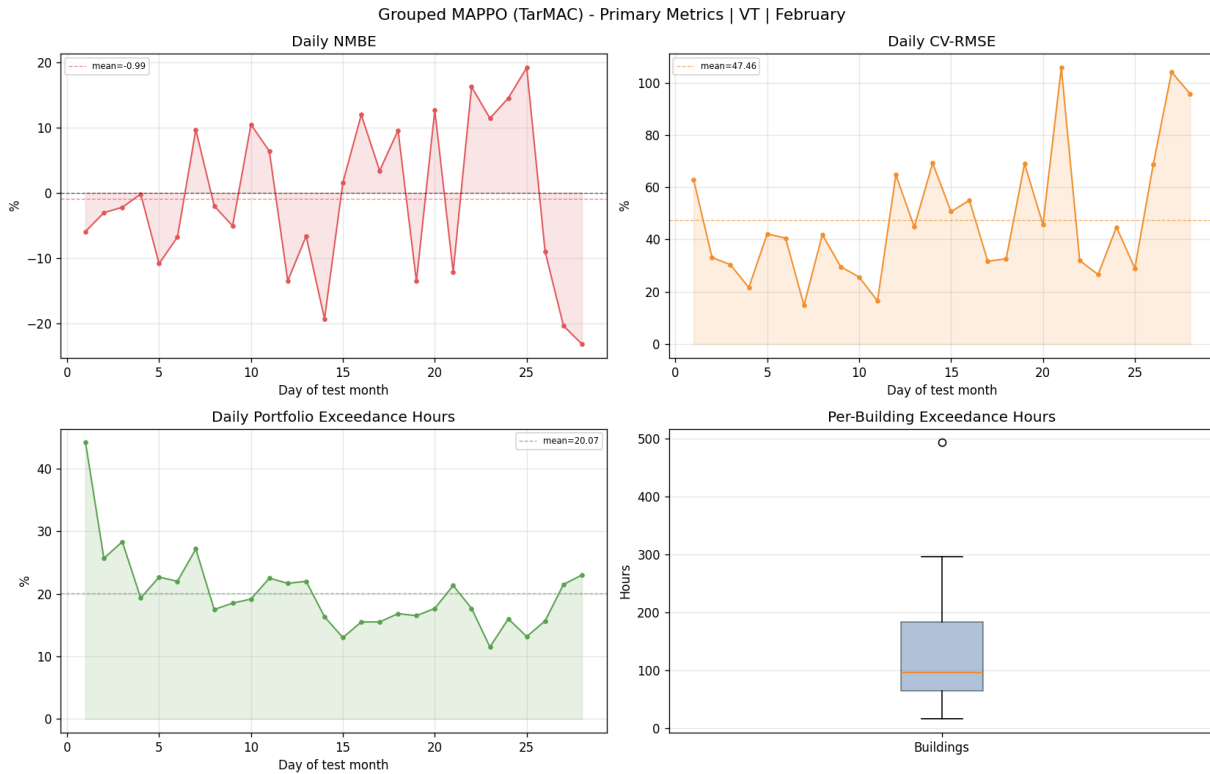


Figure 4.4: Daily primary metrics for the selected fixed 3fA controller, seed 42.

Figures 4.4 and 4.5 make the trade-off between the two representative runs clear. The fixed 3fA run has slightly lower CV-RMSE and absolute NMBE, whereas the stable router reduces



Figure 4.5: Daily primary metrics for the final stable Soft Router, router-training seed 0.

comfort exceedance from 20.07% to 16.45% and degree-hours per building-day from 4.52 to 3.43. The router therefore improves the overall balance without winning every metric in this single-run comparison.

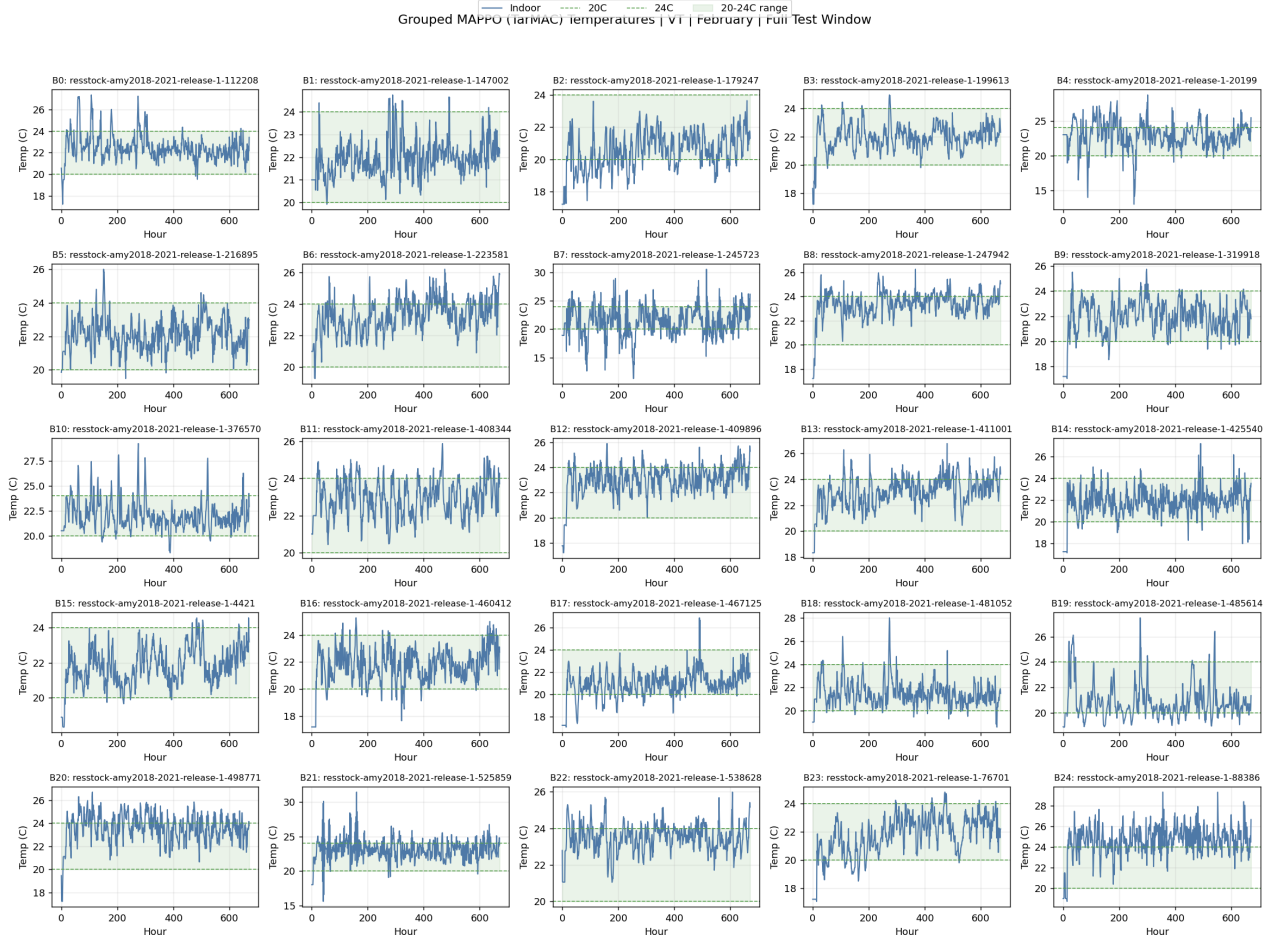


Figure 4.6: Full-month building temperatures for the selected fixed 3fA controller, seed 42.

The full-month temperature traces show that the stable router reduces the total severity of comfort violations but does not eliminate them. Several buildings still remain above the upper comfort bound for long periods, so building-level comfort remains an important limitation.

Figure 4.8 is produced in the same way as the week-1 plot in the first report: the trained checkpoint is rerun over the February test month, and the controlled portfolio load, no-storage baseline, and district target are exported for the first 168 hours.



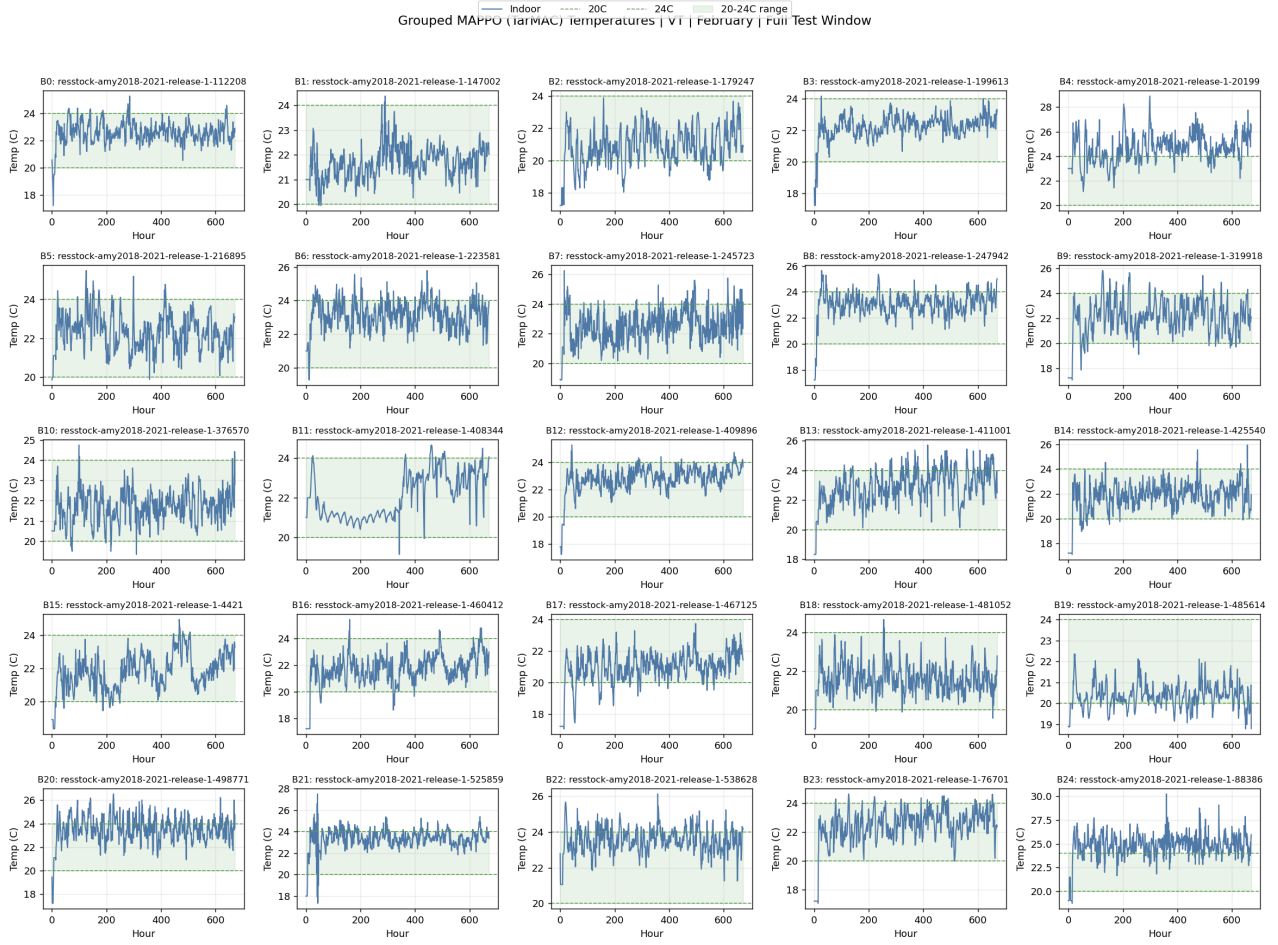


Figure 4.7: Full-month building temperatures for the final stable Soft Router, router-training seed 0.

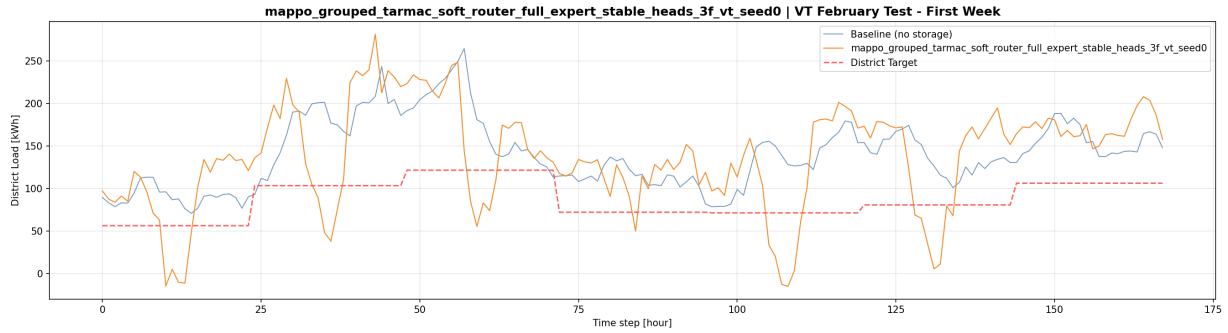


Figure 4.8: First-week aggregate load tracking for the final stable Soft Router, router-training seed 0.

## 4.9 Reward-Weight Comparison

Table 4.8 compares four reward settings using the same 3fA features, agglomerative grouping, linear-Hybrid controller, and Vermont test period.

| Setting                | $(w_{\text{nmbe}}, w_{\text{cv}}, w_{\text{comfort}}, w_{\text{binary}}, w_{\text{degree}})$ | CV-RMSE<br>(%) | NMBE <br>(%) | Degree-hours<br>(°C h/building-day) | Comfort<br>exceed. (%) |
|------------------------|----------------------------------------------------------------------------------------------|----------------|--------------|-------------------------------------|------------------------|
| Baseline               | (1, 1, 0.8, 1.3, 0.3)                                                                        | 51.93          | <b>0.49</b>  | 5.22                                | 20.27                  |
| Stricter comfort       | (1, 1, 1.5, 3.0, 1.0)                                                                        | <b>49.64</b>   | 3.04         | <b>5.10</b>                         | <b>18.74</b>           |
| Degree-hours follow-up | (1, 1, 1.5, 3.0, 1.5)                                                                        | 51.09          | 6.54         | 15.72                               | 28.83                  |
| Load-tracking focused  | (5, 5, 1.0, 1.0, 1.5)                                                                        | 50.69          | 8.32         | 22.30                               | 28.50                  |

Table 4.8: Three-seed mean comparison of the reward-weight settings in Vermont.

The mean CV-RMSE remains within a relatively narrow range of 49.64–51.93%, which allows comfort and bias to be examined while load-shape error remains at a similar level. Relative to the baseline, the stricter-comfort setting reduces CV-RMSE from 51.93% to 49.64%, degree-hours from 5.22 to 5.10, and comfort exceedance from 20.27% to 18.74%, although absolute NMBE increases from 0.49% to 3.04%. It therefore gives the best overall balance and is used for the later Vermont–Texas comparison.

The degree-hours follow-up is a direct single-parameter test: its first four weights are identical to the stricter-comfort setting, and only  $w_{\text{degree}}$  changes from 1.0 to 1.5. The higher value does not improve comfort and instead produces less stable results across seeds: the standard deviation of degree-hours increases from 1.95 to 12.77, and that of comfort exceedance increases from 1.90 to 11.93 percentage points. Increasing the load-tracking weights to 5.0 also fails to improve mean CV-RMSE and gives larger bias and comfort errors. These results demonstrate the conflict between the reward objectives and show why the largest individual weight is not necessarily the best choice.

## 4.10 Overall Interpretation and Limitations

The experiments support four conclusions. First, TarMAC Hybrid is a meaningful architectural improvement, and its advantage is best explained by compact local fusion plus a residual

communication correction. Second, grouping is part of the learning system rather than neutral preprocessing: K-means, GMM, agglomerative, and balanced spectral partitions create different accuracy. Third, using more grouping features does not necessarily improve the result. A small set of features that describe different aspects of a building is usually more effective than adding many similar features. The low bias, clear meaning, and good cross-climate performance of 3fA support this conclusion. Fourth, the Soft Router realizes the main innovation described in the Introduction. It allows each building to select different actors according to its current state, instead of always using the actor assigned by fixed grouping. This improves performance and seed stability and reduces dependence on the fixed assignment.

However, the Soft Router does not remove this dependence completely. The end-to-end and shared-representation versions perform poorly when trained from scratch; stable results are obtained only after loading strong pretrained experts. The Soft Router is therefore an improvement built on good fixed grouping: it reduces the restriction of permanently assigning each building to one actor, but the quality of the initial experts still affects the final performance.

# Chapter 5

## Conclusion and Work Plan

This report continued the first grouped-MAPPO study by asking how communication, building grouping, and grouping features should be designed together for the CityLearn portfolio load-tracking task, and whether the controller can reduce its dependence on fixed grouping. The experiments show that all three design choices affect the final policy. The strongest overall approach combines a clearer TarMAC communication structure, compact building features, agglomerative grouping, and a stable Soft Router.

### 5.1 Main Findings

First, TarMAC Hybrid improves the original TarMAC design by compressing each agent’s local information, retaining the more useful parts with ReLU, combining them with received communication, and preserving the original information through a residual connection. In the matched benchmark, it reduces CV-RMSE from 49.65% to 47.01%, comfort exceedance from 22.10% to 17.99%, and the reward magnitude from 5996 to 4715. The linear and gated ablations confirm that the improvement does not come from simply adding another layer.

Second, building grouping is part of the learning system rather than a neutral preprocessing step. The four tested methods produce different results: agglomerative clustering gives the strongest load tracking in the method comparison, so it was selected for the later feature experiments.

Third, more grouping features do not necessarily improve performance. The 3fA combination—

battery capacity, mean heating demand, and mean non-shiftable load—provides the best overall balance because it has low and stable bias, clear physical meaning, and can be adapted from Vermont heating to Texas cooling. Some other three-feature combinations are better on individual metrics, but 3fA is the most suitable overall choice. Its results also show that selecting a few features with different physical roles is more useful than adding many similar or strongly related descriptions.

Finally, the stable-head Soft Router gives the strongest routing result. Across three router-training seeds, it achieves CV-RMSE  $45.12 \pm 0.35\%$ , absolute NMBE  $1.35 \pm 0.33\%$ , degree-hours  $3.78 \pm 0.44$  per building-day, and comfort exceedance  $17.30 \pm 1.33\%$ . Its mean load tracking and comfort severity are better than those of the five-run fixed 3fA family. The staged training and anti-collapse measures also keep the router from sending almost all buildings to one actor.

## 5.2 Main Innovation

The main innovation of this work is introducing a Soft Router into grouped MAPPO to select a complete actor for each building according to its current state. Under ordinary fixed grouping, a building always uses the actor assigned to its group. With the Soft Router, the same building can use different actors at different times. The final training process first keeps the pretrained actors unchanged while learning the selector, and then fixes the learned selector while allowing only small changes to the actor output heads.

This design reduces the restriction imposed by a permanent building-to-actor assignment while retaining the useful behaviour learned by the fixed experts. The stable multi-seed results show that it is a practical performance enhancement rather than only a theoretical idea. Therefore, Soft Router can reduce the model’s dependence on fixed grouping, although it does not yet remove that dependence completely.

## 5.3 Limitations

The main limitation is that the final Soft Router still depends on strong pretrained actors. Its three reported seeds all start from the same seed-42 fixed-expert checkpoint and change

only the seed used for later router training. The results therefore show that the router can be trained consistently, but do not show that the complete model can learn equally strong experts and routing from different random initializations. The weaker end-to-end and shared-encoder experiments confirm that the current router cannot yet replace fixed grouping.

The experimental coverage is also not fully balanced. Most intermediate Soft Router variants use only seed 42 because their early results were poor and repeated training has a high time and computational cost. These runs are useful for selecting promising directions, but the final comparison still needs more strictly paired seeds.

Finally, cross-climate testing currently covers only the 25-building Vermont and Texas settings. The Texas controller is much better than the no-control and RBC references, but its load bias remains high. Some buildings also continue to violate the comfort range for long periods. The current conclusions therefore need validation under more varied conditions.

## 5.4 Future Work

The next stage should focus on strengthening the current conclusions:

- continue studying stronger grouping methods and feature combinations to improve the earlier fixed-grouping performance;
- improve Soft Router training so that it can start from weaker groups or random initialization without routing collapse;
- analyse actor-selection probabilities and building-level comfort in more detail, and further improve the current method under different climate conditions.

These experiments will determine whether Soft Router can progress from a stable enhancement of fixed grouping toward a more independent and adaptive grouping method.

# Bibliography

- [1] Daniel Beaglehole, David Holzmüller, Adityanarayanan Radhakrishnan, and Mikhail Belkin. xRFM: Accurate, scalable, and interpretable feature learning models for tabular data, 2025.
- [2] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- [3] Flora Charbonnier, Bei Peng, Julie Vienne, Eleni Stai, Thomas Morstyn, and Malcolm McCulloch. Centralised rehearsal of decentralised cooperation: Multi-agent reinforcement learning for the scalable coordination of residential energy flexibility. *Applied Energy*, 377:124406, 2025.
- [4] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM, 2016.
- [5] Abhishek Das, Theophile Gervet, Joshua Romoff, Dhruv Batra, Devi Parikh, Michael Rabbat, and Joelle Pineau. TarMAC: Targeted multi-agent communication. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97, pages 1538–1546. PMLR, 2019.
- [6] Chris Fraley and Adrian E. Raftery. Model-based clustering, discriminant analysis, and density estimation. *Journal of the American Statistical Association*, 97(458):611–631, 2002.
- [7] Kehua Li, Zhenjun Ma, Duane Robinson, and Jun Ma. Identification of typical building daily electricity usage profiles using gaussian mixture model-based clustering and hierarchical clustering. *Applied Energy*, 231:331–342, 2018.

- [8] James MacQueen. Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, pages 281–297. University of California Press, 1967.
- [9] Mikko I. Malinen and Pasi Franti. Balanced k-means for clustering. In *Structural, Syntactic, and Statistical Pattern Recognition*, volume 8621 of *Lecture Notes in Computer Science*, pages 32–41. Springer, 2014.
- [10] Andrew Y. Ng, Michael I. Jordan, and Yair Weiss. On spectral clustering: Analysis and an algorithm. In *Advances in Neural Information Processing Systems*, volume 14, 2001.
- [11] Kingsley Nweye, Kathryn Kaspar, Giacomo Buscemi, Tiago Fonseca, Giuseppe Pinto, Dipanjan Ghose, Satvik Duddukuru, Pavani Pratapa, Han Li, Javad Mohammadi, Luis Lino Ferreira, Tianzhen Hong, Mohamed Ouf, Alfonso Capozzoli, and Zoltan Nagy. Citylearn v2: Energy-flexible, resilient, occupant-centric, and carbon-aware management of grid-interactive communities. *Journal of Building Performance Simulation*, pages 1–22, 2024.
- [12] Johan Samir Obando Ceron, Ghada Sokar, Timon Willi, Clare Lyle, Jesse Farebrother, Jakob Nicolaus Foerster, Gintare Karolina Dziugaite, Doina Precup, and Pablo Samuel Castro. Mixtures of experts unlock parameter scaling for deep reinforcement learning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 38520–38540. PMLR, 2024.
- [13] Hanchuan Peng, Fuhui Long, and Chris Ding. Feature selection based on mutual information: Criteria of max-dependency, max-relevance, and min-redundancy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(8):1226–1238, 2005.
- [14] Joan Puigcerver, Carlos Riquelme Ruiz, Basil Mustafa, and Neil Houlsby. From sparse to soft mixtures of experts. In *International Conference on Learning Representations*, 2024.
- [15] Yiyun Sun, Senlin Zhang, Meiqin Liu, Ronghao Zheng, and Shanling Dong. Energy management based on safe multi-agent reinforcement learning for smart buildings in distribution networks. *Energy and Buildings*, 318:114410, 2024.



- [16] Jose R. Vazquez-Canteli, Sourav Dey, Gregor Henze, and Zoltan Nagy. Citylearn: Standardizing research in multi-agent reinforcement learning for demand response and urban energy management, 2020.
- [17] Xiangwei Wang, Peng Wang, Renke Huang, Xiuli Zhu, Javier Arroyo, and Ning Li. Safe deep reinforcement learning for building energy management. *Applied Energy*, 377:124328, 2025.
- [18] Joe H. Ward. Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association*, 58(301):236–244, 1963.
- [19] Patrick Wilk, Ning Wang, and Jie Li. Multi-agent reinforcement learning for smart community energy management. *Energies*, 17(20):5211, 2024.
- [20] Bin Zhang, Di Cao, Weihao Hu, Amer M. Y. M. Ghias, and Zhe Chen. Physics-informed multi-agent deep reinforcement learning enabled distributed voltage control for active distribution network using PV inverters. *International Journal of Electrical Power and Energy Systems*, 155:109641, 2024.